VINCENT FUNARI, PhD, CGMBS

CLINICAL BIOINFORMATICS | NGS VARIANT PIPELINES | GENOMIC INTERPRETATION

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63 PUBLICATIONS & ABSTRACTS • 4,300+ CITATIONS

SUMMARY

Clinical Bioinformatics & Genomics Executive (CSO/VP/Director-level) and Licensed Clinical Genetic Molecular Biologist Scientist with 25 years leading FDA Class III PMA, CAP/CLIA, and CE-IVD regulatory submissions across a broad assay portfolio including comprehensive genomic profiling (CGP), RNA-based gene fusion detection, CNV/indel variant calling, MSI, qPCR, NGS liquid biopsy panels, and Nanopore long-read methylation and CNV MRD. Led the first Heme NGS assay submission at NeoGenomics as VP, R&D and Bioinformatics, filed with FDA as a Class III device through the PMA pathway and carried through FDA review; the FDA review committee approved the validation proposal and draft application and described it as the "most thorough and best heme NGS proposal yet." Also supported MolDX coverage for Inivata RaDaR (MRD). Authored the submission that achieved NeoGenomics' first CGP status with MolDX, while building and scaling regulated bioinformatics infrastructure that supported growth in annual revenue from roughly $100M to over $500M (2015 to 2022). Equally deep in the reimbursement landscape, spearheaded regulatory submissions and reimbursement strategy for Medicare and New York State at EmeritusDX and Canadian-based Imagia Canexia Health (ICH) clients, laboratories across the U.S. and internationally on MolDX coverage pathways, payer strategy, and validation requirements as part of ongoing technical consulting engagements, giving equal fluency in the clinical, regulatory, and commercial sides of bringing a diagnostic assay to market.

Demonstrated academic leadership initiating the precision medicine program at Cedars-Sinai, doubling genomics revenue annually for five consecutive years and contributing to 40+ well-respected publications (Science, PNAS, AJHG, Nature Immunology, Nature Genetics, etc.), while shaping strategic business and portfolio planning, building biotechnology partnerships and advisory boards, and leading integration of wet-lab and dry-lab discovery to fuel product innovation. Equally committed to culture development. All while building transparent, high-urgency, timeline-sensitive team environments across startups and matrixed institutions, with a sustained record of recruiting, mentoring, and retaining high-performing scientific talent through periods of rapid organizational growth in genomics. Champions a culture of continuous learning and scientific development building structured training and presentation environments to cultivate the next generation of scientists, including establishing industry-leading licensing-focused presentation series that provide Continuing Education Units for licensure.

CORE COMPETENCIES

SV & CNV Domain Expertise and Independent R&D | NGS Analysis & Production Pipeline Engineering / Orchestration | Variant Interpretation, Annotation & Translation

Cross-System & LIMS Integration | Hybrid Cloud/On-Prem Infrastructure Management | AI/ML & Advanced Analytics Integration into Clinical Pipelines

Data Lifecycle & Regulatory Compliance (CAP, CLIA, NYS, FDA, CE-IVD, 21 CFR Part 11, HIPAA, MolDX) | Multi-Omics & Long-Read (Nanopore/ONT) Genomics | HPC Compute & Data Storage | GitHub Version Control

Regulatory Strategy & Reimbursement Landscape (Medicare, NYS, MolDX Coverage & Payer Pathways, CMS Billing/Coding Optimization) | Due Diligence & M&A Technical Assessment for Genomic Diagnostics

Molecular Genetics & Clinical Variant Evaluation | Executive Program & Team Leadership | Assay Development, Validation & Launch

Regulated Bioinformatics Function Build-Out (CLIA/FDA) | AWS-Based NGS/Bioinformatics Infrastructure (team leadership) | Pipeline Validation & Quality Standards | Bioinformatics Talent Development & Retention

Validation Statistics: LOD, Limit of Blank & Quantitation, Precision (Repeatability, Reproducibility), Accuracy, PPA/NPA, PPV/NPV, Linearity, ROC/AUC, Confidence Intervals, Sample-Size Justification | Deming & Linear Regression | Parametric & Non-Parametric Tests | One-Tailed Tests | Kaplan-Meier Survival Analysis | Levey-Jennings QC Plots

AI/ML & Statistical Modeling: Random Forest & Ensemble Models | Gaussian & Mixture Modeling | Clustering | k-Fold & Leave-One-Out Cross-Validation | Hyperparameter Tuning | F1 Score & Feature-Importance Analysis | Validation Against Orthogonal Truth Sets | Claude Science (proficient)

CAP/CLIA Proficiency Testing (NGS, NGS Pipelines, Molecular Assays) | Sequencing Platforms: Illumina (MiSeq, NextSeq 500, NextSeq 2000, NovaSeq) | Ion Torrent (PGM, Proton, S5, Genexus) | Element Biosciences | Oxford Nanopore (long-read methylation and CNV for MRD)

PROFESSIONAL EXPERIENCE

Private Consulting **Oct 2015 — Present**

- Advisory engagements ran alongside full-time roles through 2024.
- Conducted due diligence for mid-sized and large diagnostic companies evaluating genomic mergers and acquisitions.
- Advised diagnostic clients on market opportunities and emerging molecular and bioinformatics technologies.
- Evaluated technology and market risks in the liquid biopsy sector for venture capital firms.
- Advised US laboratories on assay validation and reimbursement strategy in the U.S. and international markets.

Principal **May 2026 — Present**

GenomicsDx, LLC — FDA/NYS/MolDX & assay validation consulting for high-complexity CLIA/CAP labs

Company A (mid-size laboratory) — wanting to add Heme NGS

- Assessed the feasibility of transferring research-use-only (RUO) assays and bioinformatics pipelines into a clinical laboratory, including reimbursement considerations.
- On-site visit to define analytical and clinical validation plans, including MolDX sample requirements. Scoped the associated budget, instruments, clinical samples, and personnel.
- Built a two-year, milestone-based implementation plan with actionable deliverables for three assays.
- Introduced a modification that saved a company $300,000 and 6 months of validation.
- Reviewed laboratory sites, instrumentation and personnel.
- Developed on-site training proposals and timelines including GAP analysis.

Company B (start-up)

- Advised on target selection and methylation assay strategy for minimal residual disease (MRD) products across the U.S., Canada, and Europe.

Company C (Heme NGS laboratory)

- Launched a new assay and bioinformatics pipeline for Element Biosciences sequencing; used hap.py benchmarking to investigate false positives associated with pipeline, polymerase, and sequencing artifacts across two laboratories.

Company D (clinical laboratory launching NGS LDTs)

- Led analytical validation readiness assessments for three NGS assays (Heme, MRD, and lymphoma clonality/monitoring), including SOP finalization and vendor training timelines, with executive reporting to leadership.
- Reviewed assay vendor contracts for pricing transparency, cybersecurity and remote access, and change-notification terms, and compared vendor error-correction sequencing methods with duplex sequencing approaches.
- Advised on Anti-Kickback Statute and Stark Law compliance for manufacturer-laboratory revenue sharing, recommending fixed per-test fees over percentage-of-revenue structures.

Principal, additional work

- Built a CMS ADLT application checklist and researched ADLT and MolDX Z-code timing and novelty argumentation.
- Analyzed FDA ASR versus IVD pathways for reagent and kit sales to clinical laboratories, and NYS CLEP licensure requirements for somatic NGS cancer testing.
- Benchmarked NGS variant pipelines against GIAB truth sets (HG002/HG003/HG004 trio): concordance, artifact profiling, and Mendelian inheritance analysis with custom Python workflows.

Head of Bioinformatics (Corvidane)

- Analyzed gene-expression data from more than 700 samples to identify treatment-associated genes and pathways for a MASH and cholesterol metabolism drug study; established QC cutoffs, assessed outliers, and validated findings.
- Compared drug-study gene-expression profiles with public GEO datasets to investigate mechanisms of action.

Chief Scientific Officer **Oct 2023 — Apr 2024**

EmeritusDX Lake Forest, United States

- Spearheaded regulatory submissions and reimbursement strategies for Medicare and New York State, successfully navigating complex requirements across 45 states to ensure compliance and optimize reimbursement outcomes while engineering a new molecular laboratory adhering to CAP guidelines, implementing rigorous quality control measures and industry best practices (with clear stage-gated program management and budgetary responsibilities) including developing templates for document control to ensure high standards of diagnostic accuracy and reliability.
- Pioneered the development and market introduction of innovative urine-based RNA/DNA liquid biopsy tests, enhancing early detection capabilities for genitourinary cancers (e.g. Bladder and Prostate Cancer) through disciplined assay launches and integration with digital reporting for clinicians.
- Enhanced and expanded pathogen panels, including STI, UTI, and GI panels, utilizing proprietary methods to achieve 5 copies per aliquot.
- Small company footprint, so wore many hats including participating in marketing discussions, trademarking, patent searches, sales visits, hands-on lab work and training for junior scientists.
- Led strategic collaborations with AccuStem pharma to clinically validate cancer “stemness” biomarkers, focusing on demonstrating clinical utility in breast cancer diagnostics while coordinating data operations and integration with downstream pipelines to support publication and IP strategy.
- Delivered impactful presentation on innovative liquid biopsy at the Association for Molecular Pathology, enhancing the company's visibility and thought leadership in the field and engaging doctors and investors on translational milestones.
- Conducted training sessions for the sales team at the National Sales Meeting and responsible for assisting in customer engagement and marketing.
- Implemented Linux server infrastructure to support Next-Generation Sequencing (NGS) and enhancing computational capabilities for the AccuStem breast cancer stemness project, integrating data systems for scalable interpretation and production hand-off.
- Procured and set up ddPCR, NGS, and quality assurance instruments, enhancing laboratory capabilities for advanced molecular diagnostics while aligning capital plans to program milestones and team development objectives.
- Supported CAP/CLIA proficiency testing, including NGS and NGS pipeline proficiency testing and other molecular assays.

Chief Scientific Officer **Jul 2022 — Sep 2023**
Imagia Canexia Health Vancouver, Canada (Remote)

- Enhanced MRD, liquid biopsy, and solid tumor DNA/RNA products, successfully closing market gaps through coordinated assay launches and integration with production data systems.
- Secured CE-IVD mark for bioinformatics pipelines, cloud and reporting infrastructures to exceed compliance and market standards, establishing documentation for IP and publication-ready validation.
- **Nanopore MRD program (Canada):** Oversaw new clinical trials on MRD testing in lung cancer patients, using Nanopore long-read sequencing for methylome profiling and CNV/structural genome detection, and coordinated data operations and analytical interpretation across sites.
- Assisted CAP/CLIA labs in Mexico, Canada, USA, Philippines, Australia, and the Middle East with validation plans and regulatory submissions supporting business objectives as a decentralized laboratory with cloud-based pipelines, marketing materials, reagent lines, customer support, etc., ensuring LIMS integration and digital QC release for scale.
- Facilitated cross-institutional partnerships to advance AI-driven standardization of MRI feature detection, allowing algorithm training and development across different datasets with Quebec's leading academic medical centers.
- Directed a 70-member cross-functional team in R&D, bioinformatics, IT and AI, enhancing clinical operations and driving innovation across global divisions by instituting accountability frameworks and structured team development.
- Led initiatives to transform organizational culture (e.g. incentive-based annual review, supported team needs, resolving silos, cross-departmental education sessions), resulting in overall company-wide departmental collaboration and efficiency.
- Developed and delivered a comprehensive strategic roadmap to the Board of Directors, aligning R&D initiatives with business objectives for new innovation and market competitiveness, in addition to presenting research at multiple scientific conferences to increase company visibility, with integrated program management and quarterly milestones mapped to product releases.
- Managed IT, AI, Computational Biology, Bioinformatics, and Molecular Departments while mentoring scientists and operators to build accountability and team development culture across programs.
- Resolved contamination issues that plagued the labs and developed new relationships with new outside vendors, instituting corrective action milestones and integrating digital monitoring to prevent recurrence.
- Supported CAP/CLIA proficiency testing, including NGS and NGS pipeline proficiency testing and other molecular assays.

Chief Scientific Officer, Clinical Division **May 2021 — Jul 2022**
NeoGenomics, Inc. Aliso Viejo, United States

- Completely revamped Project Management in organization to be more efficient and strategic while still being inclusive.
- Supported and advised on SAB with Inivata MRD clinical trials and supported US reimbursement (MolDX) application.
- Increased revenue in a single test by $7M/year through strategic billing/marking optimization.
- Developed and mentored 7 different teams of 3–5 scientists, from directors to associate scientists, in bioinformatics (pharma and clinical), regulatory, biorepository, reimbursement, and assay development (heme and solid tumor) teams; published in Human Pathology and Annals of Oncology, fostering team development and accountability.
- Spearheaded medical and scientific initiatives as a key member of the Executive Leadership Team, delivering strategic presentations to the Board of Directors and Scientific Advisory Board.
- Assumed responsibility for interdisciplinary teams for implementing 21 CFR Part 11 compliance for Bioinformatics and reporting (both FDA and Pharma initiatives), integrating data systems and program management for audit readiness.
- Optimized clinical operations by reducing input requirements for Solid Tumor NGS by 60% and shifted all molecular operations to TNA to reduce complexity/sample, improving TAT and input requirements, coordinating integration with lab workflows and LIMS.
- Continuing to migrate and improve our Cloud/On-Prem compute environment for minimum turn-around time for clinical assays, exploring GPU/Parabricks/rewriting code (e.g. improved NGS computational time for NGS solid tumor algorithm by 50%), aligning data operations with pipeline milestones.
- Launched first-ever CAP/CLIA-validated integrated complete genomic profile as ALL Heme malignancy NGS panel launched with CNV, SNP/InDels, RNA expression, and RNA fusions under CAP/CLIA by establishing a new analytical validation template for NGS-based RNA expression profile, coordinating publication planning and IP disclosures.
- >6 abstracts accepted at major oncology conferences, including ASH, AACR, ASCO, SITC, etc.; 2 research papers (TMB consortium and NTRK) accepted.
- Supported CAP/CLIA proficiency testing, including NGS and NGS pipeline proficiency testing and other molecular assays.

Vice President, R&D and Bioinformatics **Jan 2019 — Jan 2021**
NeoGenomics, Inc. United States

- Oversaw development of new LDT assays and bioinformatics during growth that led to massive increase in stock valuations ($12 → $50), instituting a single integrated roadmap across assays, pipelines, automation, and data operations with quarterly milestones.
- Scaled lab automation and integrated machine learning into MSI, CNV, and fusion pipelines.
- **Gene fusion machine learning algorithm:** Supervised the lead scientist and co-developed a random-forest ensemble in R (randomForest, caret) that combines five fusion callers (Arriba, STAR-Fusion, FusionCatcher, deFuse, Kallisto/pizzly) to classify true gene fusions from RNA-seq, using caller agreement, breakpoint coverage balance and breakpoint-spanning read counts as features. Hyperparameters were tuned and scored with F1, feature importance was ranked, and the model was validated by 10-fold cross-validation on 598 calls confirmed by PCR, Sanger sequencing and Archer. It outperformed

an n-of-five caller-agreement rule, most of all in specificity, and showed Kallisto/pizzly added nothing, reducing compute (ASH 2020, abstract 1560).

- **RNA fusion probe design algorithm:** Senior author on an algorithm that converts the genomic breakpoints of known fusions into RNA breakpoints and probe sequences by scoring candidate isoforms on database support, review status and sequence identity. Targeted capture sets for 524 and 1,632 known fusions gave 5- to 20-fold more supporting reads than whole-exome sequencing in a 12-fusion SeraSeq control, and 10- to 30-fold more in 10 clinical samples with fusions confirmed by orthogonal methods (AACR 2021, abstract 241).
- Developed and validated algorithms for heme CNV and MSI calling: an ensemble CNV algorithm tuned against FISH on thousands of clinical samples (ASH 2021) and improved MSI detection, trained and validated on NGS data against one to three orthogonal truth methods (PCR, Sanger sequencing, Archer, FISH).
- Launched a multimodal heme genomic profiling assay (DNA and RNA from total nucleic acid in a single tube) detecting SNVs, indels, CNVs, gene fusions including BCR::ABL1, and RNA expression, with 97.7% SNV/indel concordance, 7 of 7 concordant BCR::ABL1 results above 6.4% IS, and 92.5% fusion concordance against independent assays (ASH 2020).
- Lead development of LymphoTrack Clonality and MRD assays to high throughput short read sequencing.
- Reviewed and reported on new laboratory migration.
- Drove launch of multiple AI-enhanced NGS assays and liquid biopsy platforms, coordinating end-to-end integration with computational pipelines and LIMS for robust releases.
- Responsible for completely revamping SAB to achieve more focused scientific guidance and feedback.
- Reviewed and reported on scientific newly developed RNA-Seq gene fusion test including MET Exon 14 and EGFRvIII skipping events that decreased tissue usage while increasing biomarker testing with limiting tissue; this included engineering novel bioinformatic programs developed to identify RNA fusion gene probes to overcome problematic industry gaps. NYS approval for NGS gene fusion assay in record three months, supported by truth-set benchmarking and analytical interpretation.
- Initiated the development of a new biorepository department supported by three scientists enabling enterprise-wide collaborations with an estimated initial value of ~$10–20 million while also updating NGS storage requirements with IT, Medical, and Quality to optimize long-term vs short-term storage and access, integrating data systems and budgetary responsibilities.
- The FDA review committee approved the validation proposal and draft application for the first candidate Heme NGS assay, submitted as a Class III device, and described the submission as the "Most thorough and best heme NGS proposal yet." Led 21 CFR Part 11 software and systems compliance for FDA, design control inputs and outputs, validation plans and report, coordinating IP strategy and publication readiness.
- Rapid development of molecular infectious disease department; launched five highly automated COVID-19 assays including one novel highly multiplexed nanofluidic high-throughput qPCR assay (50K samples per day) that targeted all respiratory diseases and resulted in an increased $8 million revenue a month, integrating operators and doctors into digital result delivery.
- Created a 3-scientist automation team across multiple acquisitions and scaled lab automation using Hamilton STARlet/STAR, Thermo Fisher KingFisher, Promega MagMAX, to achieve higher throughput for scaling demand, while integrating machine learning into MSI, CNV, and fusion pipelines to increase accuracy, with clear team development and accountability.
- Launched and validated new NovaSeq v1.5 chemistry to save 30% on new NGS sequencing, reducing cost per test, coordinating milestones across validation and release.
- Launched two liquid biopsy assays based on NGS and qPCR, integrating data systems and clinical interpretation workflows to support reporting.
- Investigated several companies for acquisition; performed due diligence on three companies for liquid biopsy acquisition, assessing IP portfolios and integration risk across data operations.
- Several abstracts (>5) at major oncology conferences annually + 1 paper accepted (TMB).

Director of Clinical R&D and Bioinformatics **Jan 2018 — Jan 2019**
NeoGenomics, Inc. United States

- Guided major upgrades to sequencing tech and lab informatics to improve test quality, efficiency and reimbursement, including bioinformatics expansion and clinical validation of genomic profile markers TMB, HRD, and RNA fusions, integrating LIMS and data systems for automated QC release.
- Launched new UMI deduplication effort with UDI on the new NovaSeq and worked with Illumina on novel NovaSeq sequencing chemistry to overcome constant-region quality in library sequencing. Collaborated with Qiagen on prerelease of new MultiModal chemistry that includes both DNA and RNA-seq in same-tube chemistry to simplify operational workflows, coordinating program management and cross-functional milestones.
- Launched two liquid biopsy assays including both NGS and qPCR, integrating operators and clinical teams for sample-handling robustness.
- Launched new MSI qPCR test that is more sensitive than the FDA-approved Promega MSI kit, coordinating data operations and digital reporting to scale adoption.
- Delivered scientific due diligence on Genoptix to CMO and executives. Collaborated with leadership to establish enterprise-wide scientific career ladders and salary standards to improve enterprise-wide culture, reinforcing accountability and team development.
- Received first comprehensive genomic profiling status from MolDX for re-validated Solid Tumor NGS assay which included microsatellite instability and Tumor Mutation Burden, immediately increasing revenue 3-fold per test in 2019

while maintaining cost per test with strategic investments of new equipment, aligning publication strategy and IP protection.

Technical Director, Clinical Bioinformatics **Jan 2015 — Jan 2018**
NeoGenomics, Inc. United States

- Developed proprietary bioinformatic pipelines, installed and maintained servers and high-sensitivity tumor-only variant callers while building a novel Genomic ID algorithm to ID each patient while constructing an external-facing GUI bioinformatics platform (NEOSEQ), integrating LIMS and data systems for secure interpretation and release.
- Successful applications for NCI-MATCH, NYS submissions, and FDA De Novo applications for heme assays and delivered SAB and internal training presentations, coordinating publications and IP filings as part of translational milestones.
- Successfully led national scientific effort to standardize Tumor Mutation Burden for immunotherapy decisions, developed Tumor-only Homologous Recombination Deficiency (HRD) Genomic Test plus CNV, LOH and ploidy determination (CAP/CLIA certified), in addition to an Immuno-oncology Expression Profiling Panel to complement PD-L1 and TMB testing, integrating ONT proof-of-concept and long-read data interpretation where applicable.
- Co-designed and helped manage 12 clinical trials from international and national biopharma, providing program management, budgetary responsibilities, and data operations oversight through to assay launches.
- Managed evaluation, purchase and programming of liquid handlers.
- In 2016, supported some of the first NGS samples used for proficiency testing, and NGS pipeline proficiency testing, in addition to proficiency testing for other molecular assays.

Assistant Professor & Director, Genomics Core **Jan 2004 — Jan 2015**
Cedars-Sinai United States

- Spent 10 years developing precision medicine and genomics/bioinformatics lab. Responsible for every aspect of the lab from strategy, marketing, analysis, wet lab, automation, publications, grant writing, grand rounds, and teaching, to support lab operations.
- Founded Cedars' genomics platform; developed 25+ NGS/microarray/qPCR services with integrated data systems and digital access for clinicians and researchers.
- Over 25 new full ‘omic services (some only at CSMC) which include QC/QA, wet lab, bioinformatics, LIMS, storage, and reporting, integrating workflows with operators and doctors for end-to-end interpretation.
- The fastest-growing core service at CSMC, increased revenue 100%+ for five straight years. Hundreds of subscribed genomic users, supported by program management and accountable milestone tracking.
- Direct recruiting, training and supervision of seven genomic scientists (three research technicians and four bioinformaticians), emphasizing team development and clear accountability.
- Oversaw strategic planning and budgeting including purchase of new third-gen Next-Gen Sequencing Platforms, single-cell genomics enabling technologies, and managed budgetary responsibilities through to platform launches.
- Co-designed and supervised development of Clinical and Research NGS Variant Annotation and Curation pipeline and database (SiVA), integrating with LIMS and downstream pipelines for seamless hand-offs.
- Co-authored 16 grants (>$5 million), securing 8 NIH-funded grants (>$3 million).
- Discovered mutations associated with stem cell differentiation using deep exome sequencing.
- Developed first NGS assay for mycobiome profiling from scratch materials, coordinating assay launches and publication planning.
- Identified miRNAs and genes associated with Corneal Development.
- Identified key genes involved in T-Cell Development.
- Identified prognostic variant signature to differentiate benign Leiomyoma from Leiomyosarcoma.
- Characterized Exosomal RNA expression profiles in Prostate Cancer and Normal.
- Developed Single-cell and ultra-low input GEX Profiling Pipeline from extraction, library prep, sequencing, and bioinformatics, integrating data systems for scalable interpretation and benchmarking.
- Annual Lectures in Graduate Student Genomics/Genetics Series, Grand Rounds Pathology, Medical Genetics Fellowship, engaging entrepreneurs, investors, and operators on translational advances.

EDUCATION

PhD, Molecular Biology, Biochemistry and Bioinformatics, Boston University 2001
Thesis: Novel Bioinformatic Algorithms and New Molecular Approaches to Identifying Alternative Pathways in Fructose Metabolism and Hereditary Fructose Intolerance

Postdoctoral Fellowship in Genomics, Stanford School of Medicine
Warburg effect gene expression profiles in triple negative breast cancer

BS in Biology, University of California, Los Angeles 1989 — 1994

BS in Kinesiology, University of California, Los Angeles 1994 — 1995

CERTIFICATIONS

Licensed Clinical Genetic Molecular Biologist Scientist (MTP 00000445), California 2001

AWARDS

1st IGERT Bioinformatics Training Fellowship 2020
Office of Technology Development Award for Novel Algorithm, BU Office of Technology Development 1999
Dean's Award, Thesis Presentation, BU College of Arts and Sciences 2001
Clinical Research Funds Award for Research (CREF), GCRC, Cedars-Sinai 2011

PUBLICATIONS

Scientific abstracts and publications. Featured in Nature, Science, NEJM, PNAS, Human Pathology, Annals of Oncology. Frequent presenter at ASCO, AACR, ASH, ASHG, AMP and AGBT.

ASH presentation: human genetics platform for inherited heme risk genes, using NGS data with orthogonal truth for model training and validation.

NIH, CAP, FDA, CE-IVD, and MolDX-recognized regulatory submission author.

Publications in Science, Nature Immunology, PNAS, Human Pathology, Annals of Oncology, American Journal of Human Genetics, Genomics, Oncotarget, Journal of Molecular Diagnostics. Publications list available upon request. Select publications below:

- Zane LK, et al. *A Concordance Study Among 26 NGS Laboratories Participating in the National Cancer Institute Molecular Analysis for Therapy Choice (NCI-MATCH) Clinical Trial.* **Clin Cancer Res.** 2025 Jun 4.
- Weiss LM, **Funari VA**. *NTRK fusions and Trk proteins: what are they and how to test for them.* **Hum Pathol.** 2021 Jun;112:59–69.
- Vega DM, et al. *Aligning tumor mutational burden (TMB) quantification across diagnostic platforms: phase II of the Friends of Cancer Research TMB Harmonization Project.* **Annals of Oncology.** Dec 2021;32:1626–1636.
- Merino D, McShane L, Fabrizio D, **Funari V**, et al. *Establishing guidelines to harmonize tumor mutational burden (TMB): in silico assessment of variation in TMB quantification across diagnostic platforms: phase I of the Friends of Cancer Research TMB Harmonization Project.* **J Immunother Cancer.** 2020 Mar;8(1).
- Albitar A, Townsley D, Ma W, De Dios I, **Funari V**, et al. (2018). *Prevalence of somatic mutations in patients with aplastic anemia using peripheral blood cfDNA as compared with BM.* **Leukemia**, 32(1), 227–229.
- Albitar M, et al. (2018). *Correlation of MET gene amplification and TP53 mutation with PD-L1 expression in non-small cell lung cancer.* **Oncotarget**, 9(17), 13682–13693.
- Seehus CR, et al. (2015). *Innate lymphoid cell development requires TOX-dependent generation of a common ILC progenitor.* **Nature Immunology**, 16(6), 599–608.
- Taylor-Harding B, et al. (2015). *Cyclin E1 and RTK/RAS signaling drive CDK inhibitor resistance via activation of E2F and ETS.* **Oncotarget**, 6(2), 696–714.
- Lee JH, et al. (2012). *De novo somatic mutations in components of the PI3K-AKT3-mTOR pathway cause hemimegalencephaly.* **Nature Genetics**, 44(8), 941–945.
- Smits P, Bolton A, **Funari V**, et al. *Lethal skeletal dysplasia in mice and humans lacking the golgin GMAP-210.* **New England Journal of Medicine**, 362:206–216. 2010.
- **Funari VA**, et al. *Genomic analysis in a single affected individual reveals BMPER mutation in diaphanospondylodysostosis.* **American Journal of Human Genetics.** 2010.
- Merrill A, Merriman B, Farrington-Rock C, Camacho N, Sebald E, **Funari V**, et al. *Ciliary abnormalities due to defects in the retrograde transport protein DYNC2H1 in short rib polydactyly syndrome.* **Am J Hum Genet** 84:542–549. 2009.
- Tompson SW, Merriman B, **Funari VA**, et al. *A recessive skeletal dysplasia, SEMD aggrecan type, results from a missense mutation affecting the C-type lectin domain of aggrecan.* **Am J Hum Genet** 84:72–79. 2009.
- Lee H, O'Connor B, Merriman B, **Funari VA**, et al. *Improving the efficiency of genomic loci capture using oligonucleotide arrays for high throughput resequencing.* **BMC Genomics**, 10(1):646. 2009.
- Day A, Dong J, **Funari VA**, et al. *Gene characterization through large-scale co-expression analysis.* **PLOS One**, 2009.
- Rock MJ, Prenen J, **Funari VA**, et al. *Gain-of-function mutations in TRPV4 cause autosomal dominant brachyolmia.* **Nature Genetics** 40:999–1003. 2008.
- **Funari VA**, et al. *Cartilage-selective genes identified in genome-scale analysis of non-cartilage and cartilage gene expression.* **BMC Genomics** 8:165. 2007.

Select ASH and AACR abstracts

- Thomas BB, Mou Y, Keeler L, Magnan C, **Funari V** (senior scientist), Weiss L, Brown S, Agersborg S. *A Highly Sensitive and Specific Gene Fusion Algorithm Based on Multiple Fusion Callers and an Ensemble Machine Learning Approach.* **ASH 2020**, abstract 1560.
- Magnan CN, Lau-Rivera S, Lopez Diaz F, Ou C, Thomas B, Nam H, Weiss LM, Jung SC, **Funari VA** (senior author). *An Efficient Probe Design Algorithm for Direct Fusion Targeting from RNA.* **AACR 2021**, abstract 241, Cancer Research; poster 2640.
- Yu Y, et al., Ye F, **Funari V** (senior author). *A Novel Multimodal Next Generation Sequencing Assay with Total Nucleic Acid Input to Provide Comprehensive Genomic Profiling for Hematologic Malignancies.* **ASH 2020**, abstract 3379.
- Nam H, Magnan C, Lopez-Diaz F, Bender R, Agersborg S, Jung SC, **Funari V** (senior author). *Accurate Prediction of Chromosome-Level CNVs from Targeted NGS.* **ASH 2021**, abstract 3994.
- Ghosal A, et al., Ye F, **Funari VA** (senior author). *Detection of CCND1 Overexpression by RNA-Seq from TNA Samples as a Surrogate for t(11;14) Translocation in Multiple Myeloma.* **ASH**, poster 2961.